DATABASE MANAGEMENT SYSTEMS

BECS 22243

PREPARATION 02

EC/2021/005 - THAMARA G P D

1.

```
SQL> SELECT P.PNO, LISTAGG(S.CITY, ', ') WITHIN GROUP (ORDER BY S.CITY) AS Cities
  2  FROM SPJ SPJ
  3  JOIN Supplier S ON SPJ.SNO = S.SNO
  4  JOIN Part P ON SPJ.PNO = P.PNO
  5  GROUP BY P.PNO;

PN
--
CITIES
--------------------------------------------------------------------------------
P1
COLOMBO, JAFFNA

P2
COLOMBO, COLOMBO, COLOMBO, JAFFNA, JAFFNA

P3
JAFFNA, KANDY, KANDY, KANDY, KANDY, KANDY, KANDY, KANDY


PN
--
CITIES
--------------------------------------------------------------------------------
P4
JAFFNA, KANDY

P5
JAFFNA, JAFFNA, JAFFNA, KANDY

P6
JAFFNA


6 rows selected.
```

2.

```
SQL> SELECT P.PNAME, J.JNAME
  2  FROM Part P
  3  JOIN Project J ON P.CITY = J.CITY;

PNAME      JNAME
---------- --------------------
CAM        SORTER
BOLT       SORTER
COG        COLLATOR
SCREW      COLLATOR
NUT        COLLATOR
COG        TAPE
SCREW      TAPE
NUT        TAPE

8 rows selected.
```

3.

```
SQL> SELECT S.*, P.*
  2  FROM Supplier S
  3  JOIN Part P ON S.CITY = P.CITY
  4  WHERE S.STATUS >= 20;

SN SNAME                    STATUS CITY            PN PNAME           PRICE
-- -------------------- ---------- --------------- -- ---------- ----------
COLOR          WEIGHT CITY
---------- ---------- ---------------
S4 GAMINI                       20 COLOMBO         P1 NUT                 20
RED                12 COLOMBO

S1 SAMAN                        20 COLOMBO         P1 NUT                 20
RED                12 COLOMBO

S3 ANIL                         30 KANDY           P2 BOLT                14
GREEN              17 KANDY


SN SNAME                    STATUS CITY            PN PNAME           PRICE
-- -------------------- ---------- --------------- -- ---------- ----------
COLOR          WEIGHT CITY
---------- ---------- ---------------
S4 GAMINI                       20 COLOMBO         P4 SCREW               11
RED                14 COLOMBO

S1 SAMAN                        20 COLOMBO         P4 SCREW               11
RED                14 COLOMBO

S3 ANIL                         30 KANDY           P5 CAM                150
BLUE               12 KANDY


SN SNAME                    STATUS CITY            PN PNAME           PRICE
-- -------------------- ---------- --------------- -- ---------- ----------
COLOR          WEIGHT CITY
---------- ---------- ---------------
S4 GAMINI                       20 COLOMBO         P6 COG                 12
RED                19 COLOMBO

S1 SAMAN                        20 COLOMBO         P6 COG                 12
RED                19 COLOMBO


8 rows selected.
```

4.

```
SQL> SELECT S.*, SPJ.*
  2  FROM Supplier S
  3  JOIN SPJ SPJ ON S.SNO = SPJ.SNO
  4  WHERE S.STATUS < 20 AND SPJ.QTY > 200;

SN SNAME                    STATUS CITY            SN PN JN        QTY
-- -------------------- ---------- --------------- -- -- -- ----------
S2 JAGATH                       10 KANDY           S2 P3 J4        500
S2 JAGATH                       10 KANDY           S2 P3 J5        600
S2 JAGATH                       10 KANDY           S2 P3 J6        400
S2 JAGATH                       10 KANDY           S2 P3 J7        800
```

5.

```
SQL> SELECT SPJ.SNO, SPJ.PNO, SPJ.JNO, (P.PRICE * SPJ.QTY) AS Shipment_Value
  2  FROM SPJ
  3  JOIN Part P ON SPJ.PNO = P.PNO;

SN PN JN SHIPMENT_VALUE
-- -- -- --------------
S1 P1 J1           4000
S1 P2 J4           9800
S2 P3 J2           2800
S2 P3 J3           2800
S2 P3 J4           7000
S2 P3 J5           8400
S2 P3 J6           5600
S2 P3 J7          11200
S2 P5 J2          15000
S3 P3 J3           2800
S3 P4 J4           5500

SN PN JN SHIPMENT_VALUE
-- -- -- --------------
S4 P2 J1           4200
S4 P2 J4           4200
S5 P1 J2          20000
S5 P2 J2           2800
S5 P2 J4           1400
S5 P3 J4          16800
S5 P4 J4           8800
S5 P5 J4          60000
S5 P5 J5          75000
S5 P5 J7          15000
S5 P6 J2           2400

22 rows selected.
```

6.

```
SQL> SELECT DISTINCT S.CITY AS Supplier_City, P.CITY AS Part_City
  2  FROM Supplier S
  3  JOIN SPJ SPJ ON S.SNO = SPJ.SNO
  4  JOIN Part P ON SPJ.PNO = P.PNO
  5  WHERE S.CITY <> P.CITY;

SUPPLIER_CITY   PART_CITY
--------------- ---------------
COLOMBO         KANDY
JAFFNA          MATHARA
KANDY           MATHARA
JAFFNA          COLOMBO
JAFFNA          KANDY
KANDY           COLOMBO

6 rows selected.
```

7.

```
SQL> SELECT S1.SNO, S2.SNO
  2  FROM Supplier S1, Supplier S2
  3  WHERE S1.CITY = S2.CITY AND S1.SNO < S2.SNO;

SN SN
-- --
S2 S3
S1 S4
```

8.

```
SQL> SELECT S.SNAME
  2  FROM Supplier S
  3  JOIN SPJ SPJ ON S.SNO = SPJ.SNO
  4  WHERE SPJ.PNO = 'P2';

SNAME
--------------------
SAMAN
GAMINI
GAMINI
BALA
BALA
```

9.

```
SQL> SELECT S.SNO
  2  FROM Supplier S
  3  WHERE S.CITY = (SELECT CITY FROM Supplier WHERE SNO = 'S1');

SN
--
S1
S4
```

10.

```
SQL> SELECT S.SNAME
  2  FROM Supplier S
  3  WHERE S.SNO NOT IN (SELECT SPJ.SNO FROM SPJ WHERE SPJ.PNO = 'P2');

SNAME
--------------------
JAGATH
ANIL
```

11.

```
SQL> SELECT P.PNO
  2  FROM Part P
  3  JOIN SPJ SPJ ON P.PNO = SPJ.PNO
  4  WHERE P.CITY = 'COLOMBO'
  5  GROUP BY P.PNO
  6  HAVING COUNT(SPJ.JNO) > 1;

PN
--
P4
P1
```

12.

```
SQL> SELECT DISTINCT S.SNAME
  2  FROM Supplier S
  3  JOIN SPJ SPJ ON S.SNO = SPJ.SNO
  4  JOIN Part P ON SPJ.PNO = P.PNO
  5  WHERE P.WEIGHT < 15;

SNAME
--------------------
BALA
ANIL
SAMAN
JAGATH
```

13.

```
SQL> SELECT S.SNO
  2  FROM Supplier S
  3  WHERE S.STATUS < (SELECT MAX(STATUS) FROM Supplier);

SN
--
S1
S2
S4
```

14.

```
SQL> SELECT S.SNAME
  2  FROM Supplier S
  3  JOIN SPJ SPJ ON S.SNO = SPJ.SNO
  4  WHERE SPJ.PNO = 'P2';

SNAME
--------------------
SAMAN
GAMINI
GAMINI
BALA
BALA
```

15.

```
SQL> SELECT S.SNO
  2  FROM Supplier S
  3  WHERE S.STATUS < (SELECT MAX(STATUS) FROM Supplier);

SN
--
S1
S2
S4
```

16.

```
SQL> SELECT S.*
  2  FROM Supplier S
  3  WHERE S.STATUS = (SELECT MIN(STATUS) FROM Supplier);

SN SNAME                    STATUS CITY
-- -------------------- ---------- ---------------
S2 JAGATH                       10 KANDY
```

17.

```
SQL> SELECT SPJ.PNO
  2  FROM SPJ
  3  GROUP BY SPJ.PNO
  4  HAVING MIN(SPJ.QTY) > 100;

PN
--
P4
P1
P3
P6
```

18.

```
SQL> SELECT S.SNAME
  2  FROM Supplier S
  3  WHERE EXISTS (SELECT 1 FROM SPJ SPJ WHERE S.SNO = SPJ.SNO AND SPJ.PNO = 'P2');

SNAME
--------------------
SAMAN
GAMINI
BALA
```

19.

```
SQL> SELECT S.SNAME
  2  FROM Supplier S
  3  WHERE NOT EXISTS (SELECT 1 FROM SPJ SPJ WHERE S.SNO = SPJ.SNO AND SPJ.PNO = 'P2');

SNAME
--------------------
JAGATH
ANIL
```

20.

```
SQL> SELECT S.SNAME
  2  FROM Supplier S
  3  WHERE NOT EXISTS (
  4      SELECT P.PNO
  5      FROM Part P
  6      WHERE NOT EXISTS (
  7          SELECT 1
  8          FROM SPJ SPJ
  9          WHERE SPJ.SNO = S.SNO AND SPJ.PNO = P.PNO
 10      )
 11  );

SNAME
--------------------
BALA
```

21.

```
SQL> SELECT S.SNAME
  2  FROM Supplier S
  3  WHERE NOT EXISTS (
  4      SELECT PNO
  5      FROM SPJ
  6      WHERE SNO = 'S1' AND PNO NOT IN (SELECT PNO FROM SPJ WHERE SNO = S.SNO)
  7  );

SNAME
-------------------
SAMAN
BALA
```

22.

```
SQL> SELECT S.SNAME
  2  FROM Supplier S
  3  WHERE EXISTS (
  4      SELECT 1
  5      FROM Part P
  6      WHERE NOT EXISTS (
  7          SELECT 1
  8          FROM SPJ SPJ
  9          WHERE SPJ.SNO = S.SNO AND SPJ.PNO = P.PNO
 10      )
 11  );

SNAME
-------------------
SAMAN
JAGATH
ANIL
GAMINI
```

23.

```
SQL> SELECT P.PNO
  2  FROM Part P
  3  WHERE P.WEIGHT > 18
  4  UNION
  5  SELECT SPJ.PNO
  6  FROM SPJ SPJ
  7  WHERE SPJ.SNO = 'S2';

PN
--
P3
P5
P6
```

24.

```
SQL> SELECT * FROM (
  2      SELECT * FROM SPJ
  3      ORDER BY QTY DESC
  4  )
  5  WHERE ROWNUM <= 7;

SN PN JN        QTY
-- -- -- ----------
S5 P3 J4       1200
S5 P1 J2       1000
S2 P3 J7        800
S5 P4 J4        800
S1 P2 J4        700
S2 P3 J5        600
S2 P3 J4        500

7 rows selected.
```

25.

```
SQL> SELECT * FROM SPJ
  2  WHERE QTY > 100;

SN PN JN        QTY
-- -- -- ----------
S1 P1 J1        200
S1 P2 J4        700
S2 P3 J2        200
S2 P3 J3        200
S2 P3 J4        500
S2 P3 J5        600
S2 P3 J6        400
S2 P3 J7        800
S3 P3 J3        200
S3 P4 J4        500
S4 P2 J1        300

SN PN JN        QTY
-- -- -- ----------
S4 P2 J4        300
S5 P1 J2       1000
S5 P2 J2        200
S5 P3 J4       1200
S5 P4 J4        800
S5 P5 J4        400
S5 P5 J5        500
S5 P6 J2        200

19 rows selected.
```

26.

```
SQL> CREATE TABLE CS AS
  2  SELECT SNO, CITY
  3  FROM Supplier
  4  WHERE CITY = 'COLOMBO';

Table created.
```

27.

```
SQL> INSERT INTO CS (SNO, CITY)
  2  SELECT SNO, CITY
  3  FROM Supplier
  4  WHERE CITY = 'KANDY';

2 rows created.
```

28.

```
SQL> ALTER TABLE CS ADD (contact_no VARCHAR2(15));

Table altered.
```

29.

```
SQL> DROP TABLE CS;

Table dropped.
```

30.

```
SQL> UPDATE Part
  2  SET PRICE = 15, WEIGHT = WEIGHT + 5
  3  WHERE PNO = 'P2';

1 row updated.
```

31.

```
SQL> UPDATE Supplier
  2  SET STATUS = STATUS * 2
  3  WHERE CITY = 'COLOMBO';

2 rows updated.
```

32.

```
SQL> UPDATE SPJ
  2  SET QTY = 0
  3  WHERE SNO IN (SELECT SNO FROM Supplier WHERE CITY = 'COLOMBO');

4 rows updated.
```

33.

```
SQL> DELETE FROM Supplier WHERE SNO = 'S1';
DELETE FROM Supplier WHERE SNO = 'S1'
*
ERROR at line 1:
ORA-02292: integrity constraint (SYSTEM.SYS_C007001) violated - child record
found
```

Solution for the above problem is to manually delete the row in respective tables in the order first delete the record from SPJ then from Supplier.

```
SQL> DELETE FROM SPJ WHERE SNO = 'S1';

2 rows deleted.

SQL> DELETE FROM Supplier WHERE SNO = 'S1';

1 row deleted.
```

34.

```
SQL> DELETE FROM SPJ;

22 rows deleted.
```

35.

```
SQL> DELETE FROM SPJ
  2  WHERE SNO IN (SELECT SNO FROM Supplier WHERE CITY = 'COLOMBO');

0 rows deleted.
```

36.

```
SQL> ALTER TABLE Supplier
  2  ADD CONSTRAINT chk_status
  3  CHECK (STATUS BETWEEN 0 AND 50);

Table altered.
```

37.

```
SQL> CREATE TABLE SPJ (
  2       SNO VARCHAR2(2),
  3       PNO VARCHAR2(2),
  4       JNO VARCHAR2(2),
  5       QTY NUMBER(4),
  6       PRIMARY KEY (SNO, PNO, JNO),
  7       FOREIGN KEY (SNO) REFERENCES Supplier(SNO) ON DELETE CASCADE,
  8       FOREIGN KEY (PNO) REFERENCES Part(PNO) ON DELETE CASCADE,
  9       FOREIGN KEY (JNO) REFERENCES Project(JNO) ON DELETE CASCADE
 10  );

Table created.
```